

★ WEEK 16 · PART 3 · PRACTICE

From Comparison to a Trade, by Hand

Grade a Strategy's P&L; Size the Bet by Conviction

A signal vs buy & hold - and a bet that shrinks when unsure

Part 3's finale, with the arithmetic shown. We turn a winning forecast into a trading rule, grade its out-of-sample P&L against buy-and-hold, then size the final bet by conviction and the width of the credible band.

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PART 3 · APPLICATIONS

Two Steps, Computed by Hand

Case A: turn a signal into a strategy, grade its P&L and Sharpe on six held-out days against buy-and-hold. Case B: size the final bet by edge, conviction, and band width. Then a Claude Code lab.

What Part 3 Told Us

Now we make it concrete - with numbers

1

The deliverable is a decision

A scoreboard is not the finish line. The winning forecast has to become a trade you can defend.

2

The real test is out-of-sample P&L

Does the strategy beat buy-and-hold on data it never saw? We compute the P&L, the Sharpe, and the hit rate by hand.

3

Uncertainty sizes the final bet

Edge, conviction, and the width of the credible band together set how much to commit. We compute the size by hand.

Everything here is a calculator exercise: six held-out days, one signal, one sizing rule.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a signal or a set of inputs.
- Compute the P&L or the size by hand.
- Then read the trading decision it supports.

What you should watch

- Whether the strategy actually beats buy & hold.
- How the bet shrinks as the band widens.
- How the whole course lands on one trade.

From a Forecast to a Verdict

Does the signal actually make money?

The comparison picked a winning forecast; now it must earn its keep. We turn its directional signal into a strategy - go with the sign of the forecast - and grade its P&L over six held-out days against simply buying and holding.

The strategy rule

- Take the sign of the model's forecast.
- The strategy return is sign times the market.
- A right call captures the move; a wrong one loses it.

What we'll do by hand

- Compute the strategy return each day.
- Sum the P&L and compare to buy & hold.
- Compute the Sharpe and the hit rate.

Turning a Signal Into a Trade

Sign of the forecast times the market

How it works

- Forecast up: go long, earn the market return.
- Forecast down: go short, earn its negative.
- So strategy return = $\text{sign}(\text{signal}) \times \text{market}$.

Why grade it this way

- It is the simplest use of a directional call.
- It reveals whether the signal has real edge.
- The benchmark is always buy-and-hold.

The Held-Out Test: Six Days

Signal, market, and the strategy return

Day	signal	market (%)	strategy = sign x market
1	+ (up)	+0.8	+0.8
2	+ (up)	-0.5	-0.5
3	- (down)	-1.2	+1.2
4	+ (up)	+0.6	+0.6
5	- (down)	+0.4	-0.4
6	+ (up)	+1.0	+1.0

Goal: sum the strategy column, compare to summing the market column, then compute Sharpe and hit rate.

P&L, Sharpe, and Hit Rate

Grade the strategy against buy & hold

$$\text{strategy}_t = \text{sign}(\text{signal}) \times \text{market}_t$$

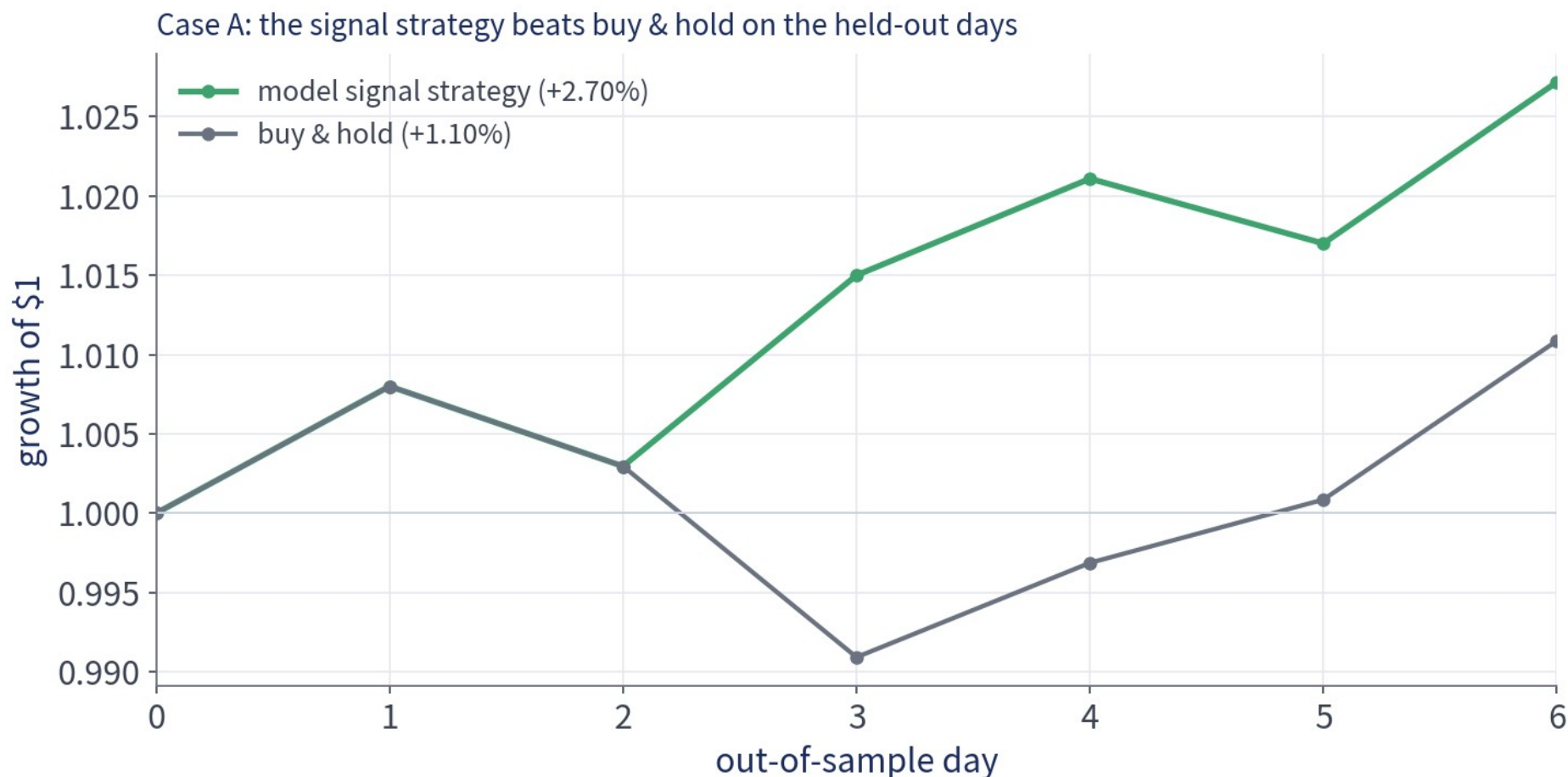
total P&L: strategy = +2.70%, buy & hold = +1.10%

Sharpe (per day): $\frac{0.45}{0.73} = \mathbf{0.62}$ vs $\frac{0.18}{0.85} = 0.21$, hit rate = $\frac{4}{6} = 67\%$

The strategy makes +2.70% versus +1.10% for buy & hold, with a higher per-day Sharpe and a 4-of-6 hit rate.

The Strategy Beats Buy & Hold

The same numbers as an equity curve. On the held-out days the signal strategy ends higher than buy-and-hold - the ultimate grade for a trading study. But six days is short, so treat it as an illustration, not proof.



The Real Verdict

Beating the benchmark, honestly

What to ask

- Did it beat buy-and-hold out of sample?
- Is the edge worth the extra complexity?
- Does it survive a longer test and costs?

What to resist

- Reading too much into six days.
- Ignoring trading costs and slippage.
- Claiming victory on a thin margin.

Size the Bet by Conviction

The final decision the whole course points to

A verdict is not enough - you must decide how much to commit. Conviction sizing multiplies the edge by the probability the setup is favorable and by how tight the credible band is. When the band is wide, the bet shrinks. We compute it by hand.

The three inputs

- Edge: the expected move if right.
- Conviction: $P(\text{favorable})$, from the regime.
- Band: how tight the forecast is.

What we'll do by hand

- Compute the base conviction size.
- Adjust it for a tight and a wide band.
- See the bet shrink as uncertainty rises.

Sizing by Conviction

Edge, conviction, and the band

The logic

- Start from the raw edge.
- Scale by how sure the regime is favorable.
- Shrink when the credible band is wide.

The behavior

- Full size only when all three agree.
- Automatic caution when the model is unsure.
- Uncertainty sets the bet, not a footnote.

Compute the Bet Size

Edge x conviction x band

$$\text{size} = \text{edge} \times P(\text{favorable}) \times (1 - \text{band})$$

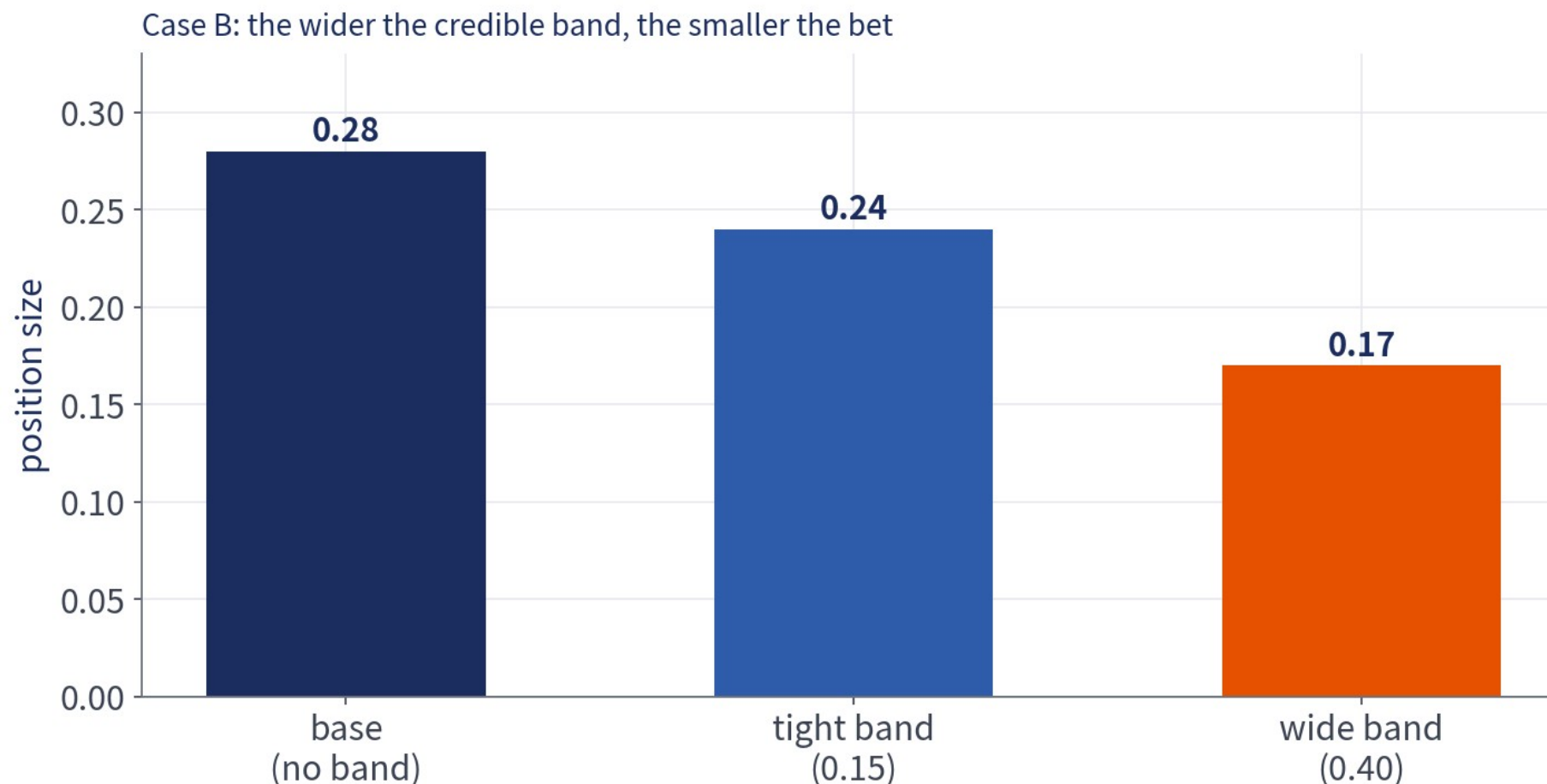
$$\text{base} = 0.40 \times 0.70 = 0.28$$

$$\text{tight band (0.15)} : 0.28 \times 0.85 = 0.24, \quad \text{wide band (0.40)} : 0.28 \times 0.60 = \mathbf{0.17}$$

With a tight band the bet is 0.24; with a wide band it falls to 0.17 - about 30% smaller, purely because the model is less sure.

A Wider Band, a Smaller Bet

The base conviction size shrinks step by step as the credible band widens. The model's honesty about its own uncertainty flows directly into how much capital you put at risk - exactly the quant-quake lesson.



What Conviction Sizing Buys

A little upside for much less regret

The trade-off

- A smaller bet gives up some expected gain.
- But it loses far less when the call is wrong.
- You commit most only when you are most sure.

Why it's worth it

- It caps the damage from an overconfident model.
- It keeps you in the game through bad stretches.
- It turns honest doubt into safer risk.

One Pair Trade, Every Model

The whole course, assembled by hand

Here is the capstone in one worked example: a single pairs trade in which every time-series model you built contributes one number. Cointegration, a z-score, OU, GARCH, an HMM, and a Bayesian band together produce a complete trade ticket - computed by hand.

Each model's job

- Cointegration: the hedge ratio and the spread.
- z-score & OU: the entry and the holding period.
- GARCH, HMM, Bayesian: the size and the gate.

The output

- A direction: short or long the spread.
- A size: the fraction of the book to risk.
- An exit and a stop, from the models.

The Spread and the Entry

Cointegration sets the pair; the z-score triggers

$$(1) \text{ Cointegration: } \beta = \frac{\text{Cov}(A, B)}{\text{Var}(B)} = \frac{0.070}{0.092} = 0.76, \quad \text{spread} = A - 0.76 B$$

$$(2) \text{ z-score today: } z = \frac{\text{spread} - \mu}{\sigma} = +2.1 \Rightarrow \text{spread is rich, SHORT it}$$

Cointegration (W7) gives the hedge ratio 0.76 and the spread; today's z-score of +2.1 says the spread is rich - so we SHORT it.

The Holding Period

OU turns mean-reversion speed into days

$$(3) \text{ OU half-life} = \frac{\ln 2}{-\ln \phi} = \frac{0.693}{-\ln 0.75} = \frac{0.693}{0.288} = \mathbf{2.4} \text{ days (holding period)}$$

What it gives

- The spread's AR(1) coefficient is 0.75.
- Its half-life is about 2.4 days.
- So we expect to hold the trade ~2 days.

Why it matters

- It sets a realistic exit horizon.
- It tells you if the edge decays too slowly.
- OU (Week 11) made this a number.

The Size and the Gate

GARCH sizes, the HMM gates, Bayes sets conviction

$$(4) \text{ GARCH size: } \frac{\text{target risk}}{\text{vol}} = \frac{0.010}{0.018} = 0.56$$

$$(5) \text{ HMM gate: } P(\text{crisis}) = 0.30 \Rightarrow \text{gate} = 1 \quad (6) \text{ conviction } 0.72, \text{ band } 0.20$$
$$\text{final size} = 0.56 \times 0.72 \times (1 - 0.20) \times 1 = \mathbf{0.32}$$

GARCH (W5-6) vol-targets to 0.56; the HMM (W10) gate is ON since $P(\text{crisis})$ is low; the Bayesian band and conviction (W15) trim it to a final size of 0.32.

One Pair Trade, Every Model

Assembled: SHORT the spread (short A, long 0.76 B), size 0.32 of the book, exit near $z = 0$ in about two days, stop out if $P(\text{crisis})$ rises above 0.5. Every number on the ticket came from a model you built.

That is what all the time series was for

One Pair Trade, Every Model

The whole course, assembled into a single decision - each model contributes one number.



The trade ticket

SHORT the spread (short A, long 0.76 B), size 0.32 of the book, expected exit near $z = 0$ in about 2 days, stop out if $P(\text{crisis})$ rises above 0.5. Every number came from a model you built this course.

That is what all the time series was for.

The Common Pattern

Two calculations, one deliverable

Both cases, computed by hand, turn analysis into action. A comparison becomes a strategy graded honestly on its P&L. A forecast becomes a bet whose size reflects real conviction. Together they are the capstone: an honest verdict, and a trade you can defend

The trap

- Stopping at a scoreboard.
- Betting the same regardless of conviction.
- Claiming an edge you never traded out of sample.

The fix

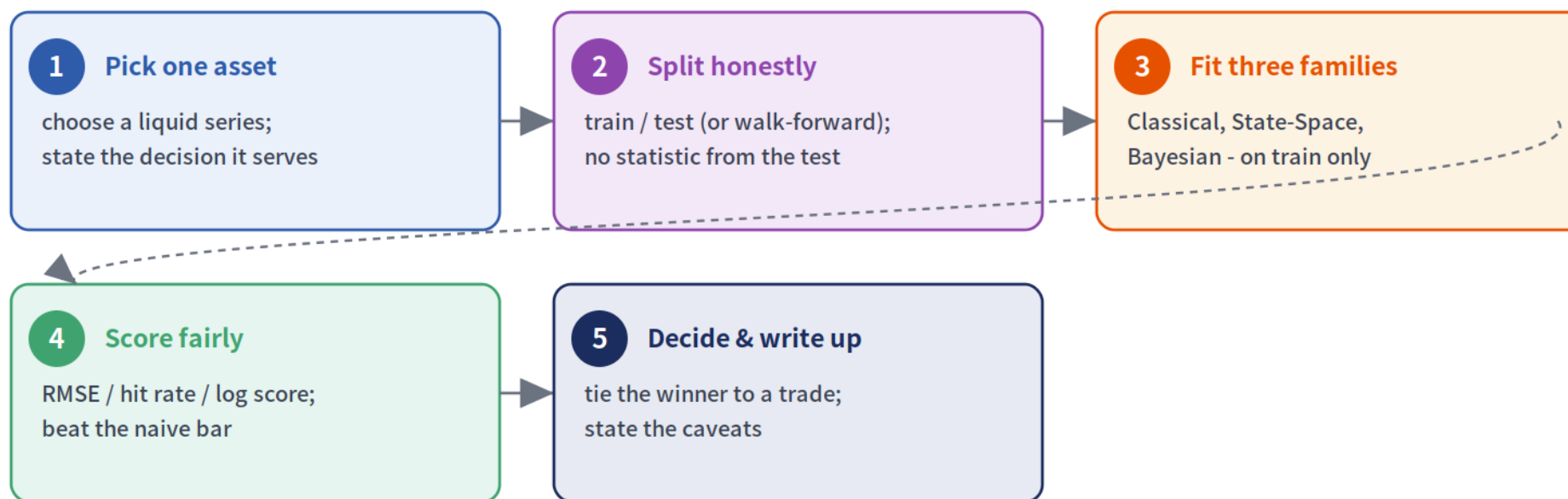
- Grade the strategy's P&L against buy & hold.
- Size the bet by edge, conviction, and band.
- Deliver a decision, with its caveats.

Where the Hand Calcs Fit

Case A is the 'decide and write up' box - grading the strategy's P&L. Case B is the sizing that turns the verdict into a bet. Together they complete the final project pipeline, from one asset to one defensible trade.

Your Final Project, Step by Step

One asset, three families, a fair test, an honest verdict - and a trading recommendation.



The deliverable is not the model - it is the honest verdict and the decision it supports.

Reproduce the Hand Calcs, Then Backtest

Direct Claude Code to check your arithmetic

GOAL Verify both steps in code, then run the full strategy on a real asset.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - grade the strategy:

- * take the 6-day signal/market table; compute strategy = sign x market; print daily P&L;
- * print total P&L, per-day Sharpe, and hit rate for strategy and buy-and-hold.

CASE B - size the bet:

- * compute size = edge x P(favorable) x (1-band);
- * print the base, tight-band, and wide-band sizes.

Then run the signal strategy on a real asset, out of sample, with costs, vs buy-and-hold.

Check against the slides

- Strategy +2.70% vs buy-and-hold +1.10%.
- Per-day Sharpe 0.62 vs 0.21.
- Hit rate $4/6 = 67\%$.
- Size: base 0.28, tight 0.24, wide 0.17.

Deliverable: `part3_practice.py` matching the hand calculations, then a real out-of-sample backtest.

Three Questions on the Cases

If you can answer these, the course has landed

Q1 How do you turn a directional forecast into a P&L?

**Strategy return =
 $\text{sign}(\text{signal}) \times \text{market}$**

Q2 What is the ultimate grade for the strategy?

**Out-of-sample P&L vs
buy & hold**

Q3 What happens to the bet when the band widens?

**It shrinks - you risk
less when unsure**

All three come from the strategy P&L, the buy-and-hold benchmark, and the conviction-sizing rule.

Three Pitfalls to Avoid

Where the final step slips

Common mistakes

- "A good scoreboard finishes the project."
- "Bet the same size no matter the conviction."
- "Six good days prove the strategy."

The reality

- Grade the strategy's out-of-sample P&L too.
- Size by edge, conviction, and band width.
- Test longer, with costs, before you believe it.

Two Calculations to Remember

Part 3 applications in one screen

1

Grade the strategy's out-of-sample P&L

Strategy = sign(signal) x market gave +2.70% versus +1.10% for buy-and-hold, with a higher per-day Sharpe and a 4-of-6 hit rate. The P&L, not the RMSE, is the final grade of a trading study.

2

Size the bet by conviction and the band

Size = edge x P(favorable) x (1 - band). A wide band cut the bet from 0.24 to 0.17 - about 30% smaller - purely because the model was less sure. Honest doubt becomes safer risk.

3

That is the whole course, in one trade

A point sets the hedge, a regime gates it, a band sizes it - all tested out of sample first. Match the model to the decision, test everything, and let uncertainty size the bet.

The core course is complete - thank you, and good luck with the final project.